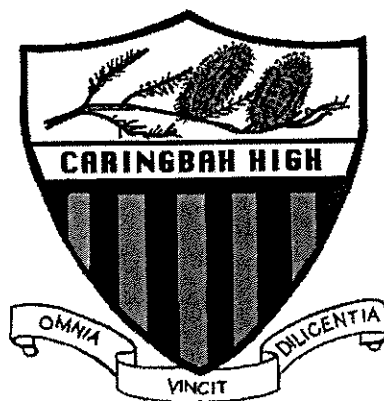




Caringbah High School

Year 12 2018

Mathematics

HSC Course

Trial HSC

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A reference sheet is provided.
- In Questions 11-16, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for partial or incomplete answers

Total marks – 100

Section I 10 marks

Attempt Questions 1-10

Mark your answers on the answer sheet provided. You may detach the sheet and write your name on it.

Section II 90 marks

Attempt Questions 11-16

Write your answers in the answer booklets provided. Ensure your name or student number is clearly visible.

Name: _____

Class: _____

Marker's Use Only								
Section I	Section II						Total	
Q 1-10	Q11	Q12	Q13	Q14	Q15	Q16		
/10	/15	/15	/15	/15	/15	/15	/100	

Section I

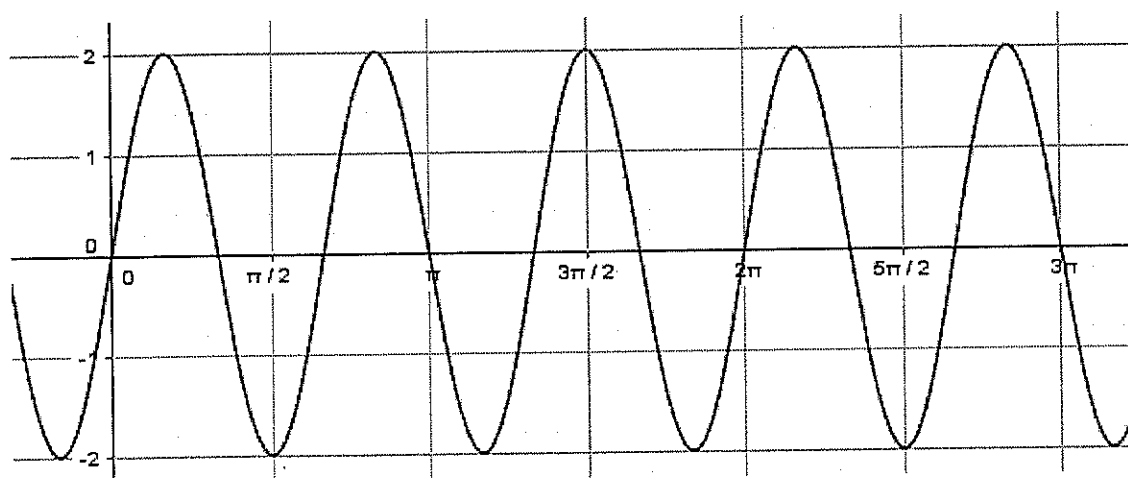
10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

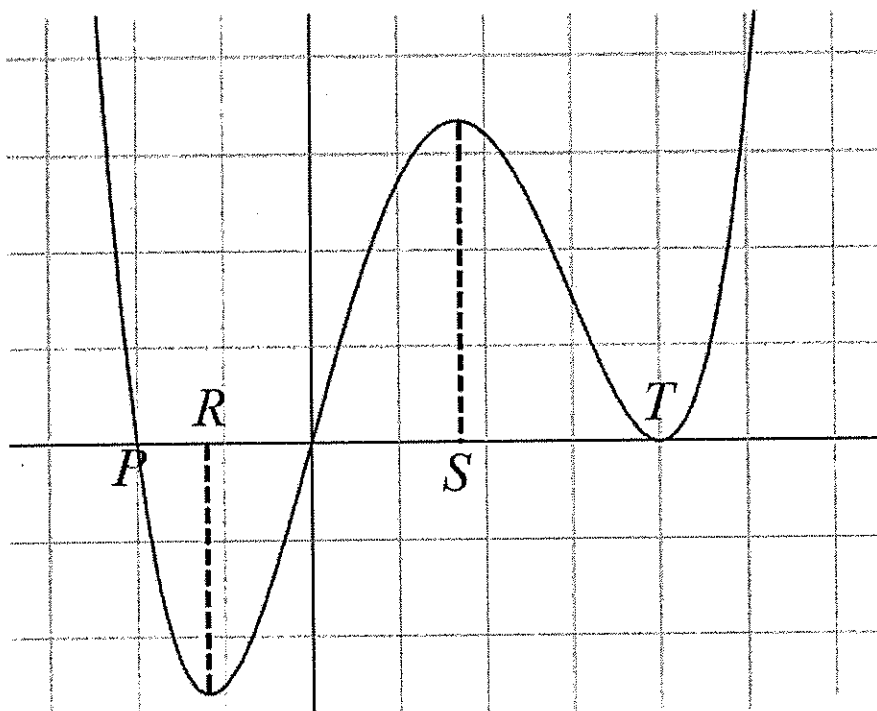
Use the multiple-choice answer sheet for Questions 1–5

1. The graph below shows the function:



- (A) $y = 3 \sin 2x$
- (B) $y = 2 \sin 3x$
- (C) $y = 3 \sin\left(\frac{x}{2}\right)$
- (D) $y = 2 \sin\left(\frac{x}{3}\right)$
2. $\int \frac{dx}{3x+1} =$
- (A) $\frac{1}{3} \log_e(3x+1) + c$
- (B) $\log_e\left(\frac{3x+1}{3}\right) + c$
- (C) $3 \log_e(3x+1) + c$
- (D) $\log_e(3x+1) + c$

3. The diagram below shows the graph of $y = f'(x)$.



Which letter indicates the position of the x -value of the maximum turning point of the function $y = f(x)$?

- (A) P
 - (B) R
 - (C) S
 - (D) T
4. A parabola has the equation $x^2 + 2x + 25 = 8y$. The focus of the parabola is:

- (A) $(-1, 1)$
- (B) $(-1, 5)$
- (C) $(1, 1)$
- (D) $(-1, 3)$

5. The fourth term of a geometric series is 192 and the seventh term is -24 .

What is the infinite sum of the series?

- (A) 1024
- (B) -128
- (C) 128
- (D) -1024

6. If $\int_2^5 f(x) dx = 7$, what is the value of $\int_2^5 1 - f(x) dx$?

- (A) -6
- (B) -4
- (C) 8
- (D) 10

7. The domain of the function $y = \log_e(2x - 1)$ is:

- (A) All real x , $x \neq \frac{1}{2}$
- (B) $x \geq \frac{1}{2}$
- (C) $x > \frac{1}{2}$
- (D) $x \leq \frac{1}{2}$

8. A circular metal plate of area $A \text{ cm}^2$ is being heated. It is given that

$$\frac{dA}{dt} = \frac{\pi t}{32} \text{ cm}^2/\text{h}$$

What is the exact area of the plate after 8 hours, if initially the plate had a radius of 6 cm?

- (A) $\pi \text{ cm}^2$
- (B) $0.25\pi \text{ cm}^2$
- (C) $36\pi \text{ cm}^2$
- (D) $37\pi \text{ cm}^2$
9. The series $1+2+4+5+7+8+\dots\dots\dots+199+200$ is obtained by omitting from the first 200 positive integers all the multiples of 3. The sum of this series is:
- (A) 20100
- (B) 10050
- (C) 17688
- (D) 13467
10. It is assumed that the number $N(t)$ of ants in a certain nest at time $t \geq 0$ is given by

$$N(t) = \frac{A}{1+e^{-t}} \text{ where } A \text{ is a constant and } t \text{ is measured in months.}$$

At time $t = 0$, $N(t)$ is estimated at 2×10^5 ants. What is the value of A ?

- (A) 2×10^5
- (B) 2×10^{-5}
- (C) 4×10^5
- (D) 4×10^{-5}

End of Section I

Section II 60 marks**Attempt Questions 11–16****Allow about 2 hours 45 minutes for this section**

Answer each question in a SEPARATE writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

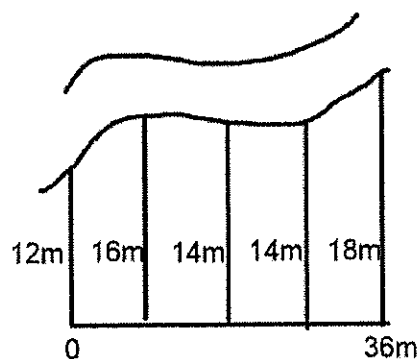
Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) Express $\frac{2}{\sqrt{7}-1}$ with a rational denominator in its simplest form. 2
- (b) Express $\log_3 9 + \log_3 27$ as an integer. 2
- (c) Evaluate $\int_0^3 e^{2x+3} dx$, expressing your answer as an exact value. 2
- (d) Differentiate the following with respect to x :
- (i) $\tan(x^2)$ 1
- (ii) $x^2 e^x$ 2
- (iii) $\frac{1}{3x^6}$ 2
- (e) Show that $\sqrt{2} + \sqrt{8} + \sqrt{18} + \dots$ is an arithmetic series and hence find the exact value of 2
the sum of the first 50 terms in simplest form.
- (f) Prove that $\frac{\sin^3 \theta}{\cos \theta} + \sin \theta \cos \theta = \tan \theta$ 2

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) The diagram below shows a paddock with one side bounded by a river.



Use Simpson's Rule with the 5 function values shown on the diagram to find the approximate area of the paddock. 1

- (b) A person walks 21 km due east from A . Another person walks 12 km from A on a bearing of $322^\circ 29'$. How far apart are they, correct to the nearest km? 2

- (c) Consider the curve $y = x^3 - 6x^2 - 9x + 4$

- (i) Find the stationary points and determine their nature. 4
- (ii) Find the coordinates of any points of inflexion. 2
- (iii) Find the values of x for which the curve is increasing and concave up. 1
- (iv) Hence draw a half page sketch of the curve $y = x^3 - 6x^2 - 9x + 4$ 2

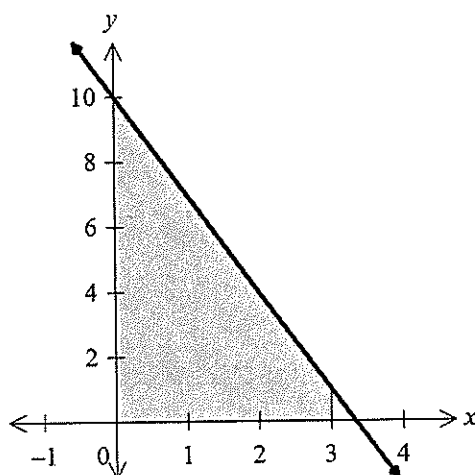
- (d) Given that a parabola has its focus at $(2,1)$ and vertex at $(3,1)$ find:

- (i) the focal length 1
- (ii) the equation of the directrix 1
- (iii) the equation of the parabola 1

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) The shaded region that lies between the x -axis and the line $y = 10 - 3x$ from $x = 0$ to $x = 3$ is rotated about the x -axis to form a solid of revolution. Find the exact volume of this solid. 3



- (b) The graph of a function has the following properties: 2

- It is continuous
- Passes through the origin and has a minimum turning point at $(4, 0)$.
- Concave up for $x > 3$ and $x < 0$.
- Increasing for $-2 < x < 2$ and $x > 4$.

Sketch a possible graph of the function.

- (c) Find:

(i) $\int \frac{dx}{\sqrt{x+6}}$ 2

(ii) $\int_0^2 \frac{x}{x^2 + 4} dx$ leaving your answer as an exact value in simplest form. 2

- (d) (i) Sketch the graph of $y = 1 + 2 \sin x$ for $0 \leq x \leq 2\pi$ 2

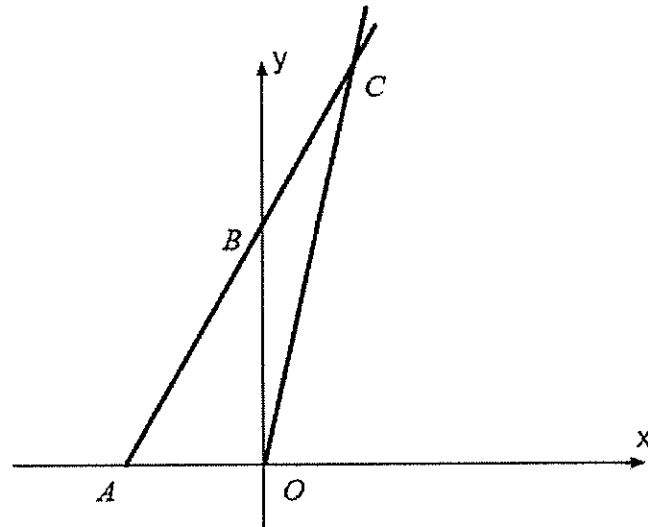
- (ii) Find the exact value of the coordinates of the points where the graph of $y = 1 + 2 \sin x$ crosses the x -axis in the domain $0 \leq x \leq 2\pi$ 2

- (e) Factorise fully $16x^2 - 16a^2 + 24a - 9$ 2

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

(a)



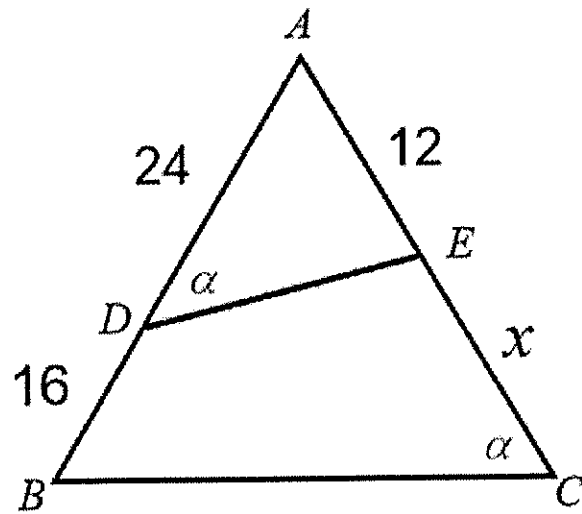
In the diagram above A and B are points on the x and y axes respectively. The line AB has equation $\sqrt{3}x - y + \sqrt{3} = 0$. The point C lies on AB such that the area of $\triangle AOC$ is $3\sqrt{3}$ square units.

- | | | |
|-------|---|---|
| (i) | Show that the coordinates of A and B are $(-1, 0)$ and $(0, \sqrt{3})$. | 1 |
| (ii) | Find the gradient of AB . | 1 |
| (iii) | Find the size of $\angle BAO$. | 1 |
| (iv) | Hence find the length AC . | 2 |
| | | |
| (b) | Find the exact value of $\tan \frac{\pi}{3} + \operatorname{cosec} \frac{\pi}{4}$ | 2 |
| | | |
| (c) | Find the values for x so that $\log_e x = 2 \log_e 2x$ | 2 |
| | | |
| (d) | Consider the quadratic function $x^2 - (k+2)x + 4 = 0$
For what values of k does the quadratic function have real roots? | 2 |

Question 14 continues on page 10

- (e) In the diagram below, $\triangle ABC \parallel \triangle ADE$.

2



Find the value of x .

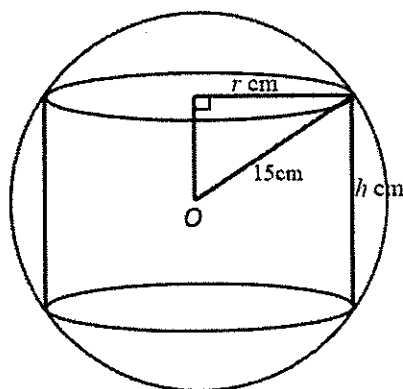
- (f) Find $\lim_{x \rightarrow 3} \frac{x^2 - x - 6}{x - 3}$

2

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

- (a) A cylindrical block of height h cm base radius r cm is cut from a solid sphere of radius 15 cm. O is the centre of the sphere.

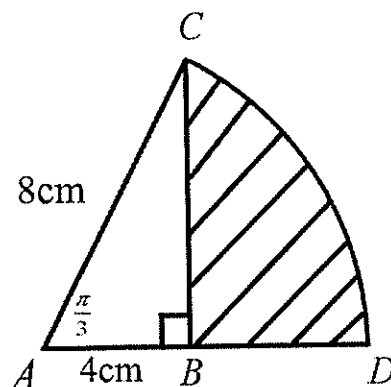


- (i) Show that the volume of the cylinder is given by $v = 225\pi h - \frac{\pi}{4}h^3$ 3
- (ii) Find the height of the cylinder of maximum volume which can be cut from the sphere. 3
- (b) If α and β are the two roots of $2x^2 + 3x - 4 = 0$, find the value of:
- (i) $\alpha + \beta$ 1
- (ii) $\alpha^3 + \beta^3$ 2
- (c) A family borrows \$120 000 to extend their house. The bank loans them the money for 10 years at an interest rate of 6% p.a. on the balance owing at the end of each month. The loan is to be repaid in equal monthly instalments of \$ M . Let A_n be the amount owing after n months.
- (i) How much do they owe at the end of the first month *before* the first payment is made? 1
- (ii) Show that at the end of the second month immediately *after* the second payment has been made the amount owing is given by 2
- $$A_2 = (121203 - 2.005M)$$
- (iii) Show that at the end of n months after n repayments have been made the amount owing will be given by 2
- $$A_n = 120\,000 \times 1.005^n - 200M(1.005^n - 1)$$
- (iv) Hence find the value of each monthly instalment \$ M . 1

End of Question 15

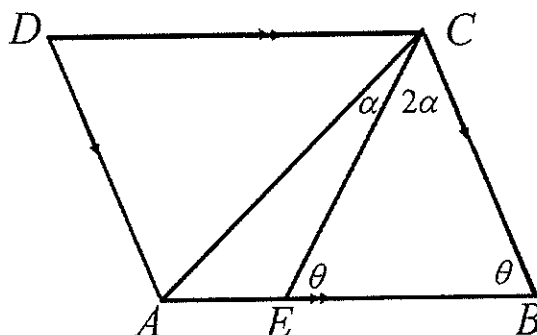
Question 16 (15 marks) Use a SEPARATE writing booklet.

- (a) Water is draining from a storage tank at a rate that is proportional to the volume of water that is left contained in the tank. Let V be the amount of water in the tank.
- (i) Show that $V = V_0 e^{-kt}$ is a solution to the differential equation $\frac{dV}{dt} = -kV$. 1
- (ii) When full, the water tank holds 1000 L of water. After 40 minutes, the tank has 800 L of water. How much water will be left in the tank after 1 hour to the nearest litre? 2
- (iii) How long in hours and minutes will it take for there to be 10 L of water left in the tank? 2
- (b) A right-angled triangle ACB is placed in a sector of a circle with radius 8 cm. 2
 $AB = 4$ cm and $\angle BAC = \frac{\pi}{3}$ radians.



Calculate the exact area of the shaded region BCD .

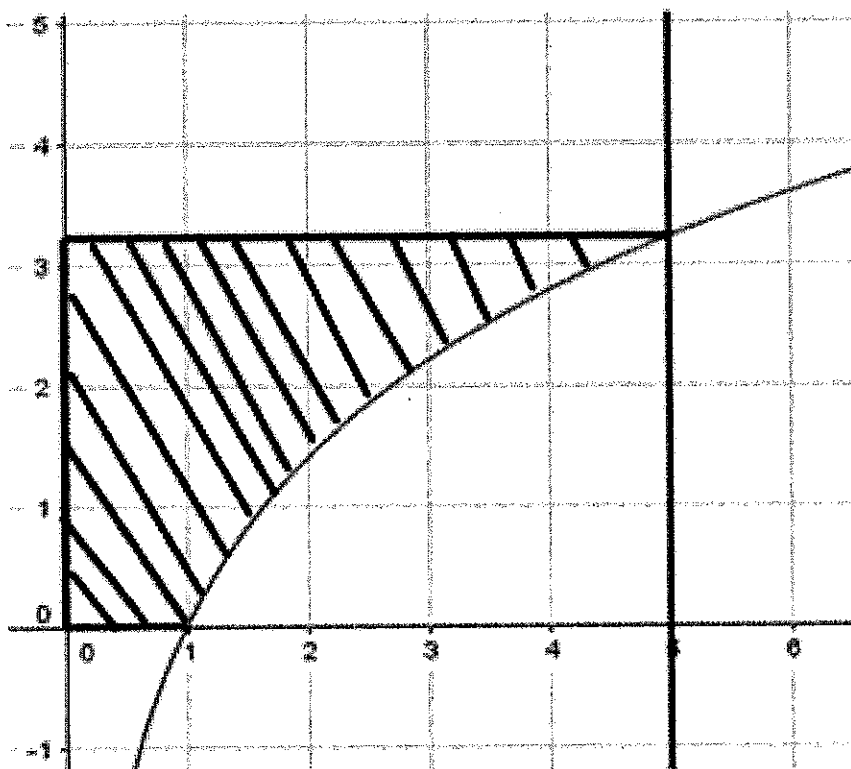
- (c) $ABCD$ is a rhombus. The angles involving α and θ are marked on the diagram. 3



Determine the size of α and θ giving clearly worded reasons.
 (Diagram is not drawn to scale.)

Question 16 continues on page 13

- (d) (i) Show that $\frac{d}{dx}(x \log_e x - x) = \log_e x$ 1
- (ii) Hence, or otherwise, find $\int \log_e(x^2) dx$ 1
- (iii) The graph shows the curve $y = \log_e(x^2)$ which meets the line $x = 5$ at Q . 3



Using your answers from (i) and (ii), or otherwise, find the exact area of the shaded section.

End of Paper



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Student Number

Examination:.....

2018 2 unit Trial CHS

1. $a = 2$ $T = \frac{2\pi}{n} = \frac{2\pi}{3}$ so $n = 3$ $y = 2 \sin 3x$ [B]

2. $\int \frac{dx}{3x+1} = \frac{1}{3} \ln(3x+1) + C$ [A]

3. $\sqrt{0} - \max P = [A]$

4. $(x+1)^2 = 8(y-3)$ vertex $(-1, 3)$ $a = 2$ focus: $(-1, 5)$ [B]

5. $T_H = ar^3 = 192$ $r^3 = \frac{-24}{192} = -\frac{1}{8}$ so $r = -\frac{1}{2}$
 $T_7 = ar^6 = -24$ $S = \frac{a}{1-r}$ $a \times -\frac{1}{8} = 192$ so $a = -1536$
 $S = \frac{-1536}{1+\frac{1}{2}} = \frac{-3072}{\frac{3}{2}} = -1024$ [D]

6. $\int_2^5 (1-f(x)) dx = x \Big|_2^5 - 7 = 3 - 7 = -4$ [B]

7. $y = \log_e(201-x)$ $D: 201-x > 0$ $x < \frac{1}{2}$ [C]

8. $\frac{dA}{dt} = \frac{\pi b}{32}$ $\therefore A = \frac{\pi x^2}{64} + C$ $t=0$ $A = 36\pi$ [D]

9. $100 \times 201 = 33 \times 201 = 67 \times 201 = 13467$ [D]

10. $N(0) = \frac{A}{1+1} = 2 \times 10^5$
 $A = 4 \times 10^5$ [C]

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Q11 a) $\frac{2}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} = \frac{2\sqrt{3}+2}{6} = \frac{\sqrt{3}+1}{3}$

b) $\log_3 9 + \log_3 27 = 2 \log_3 3 + 3 \log_3 3$
 $= 5 \times 1 = 5$

c) $\int_0^3 e^{201x} dx = \frac{1}{201} e^{201x} \Big|_0^3 = \frac{1}{201} (e^{603} - e^0)$

d) i) $\frac{d \tan(x)}{dx} = \sec^2(x)$

ii) $\frac{d x^2 e^x}{dx}$ $u = x^2$ $v = e^x$
 $u' = 2x$ $v' = e^x$
 $= x^2 e^x + 2x e^x$
 $= x e^x (x+2)$

iii) $\frac{d \frac{1}{3} x^{-6}}{dx} = -\frac{6}{3} x^{-7}$
 $= -\frac{2}{x^7}$

e) $\sqrt{2} + \sqrt{8} + \sqrt{18} = \sqrt{2} + 2\sqrt{2} + 3\sqrt{2}$

$a = \sqrt{2}$ $d = \sqrt{2}$

$S_{50} = \frac{50}{2} (2\sqrt{2} + 49\sqrt{2})$

$= 25 \times 51\sqrt{2}$

$= 1275\sqrt{2}$

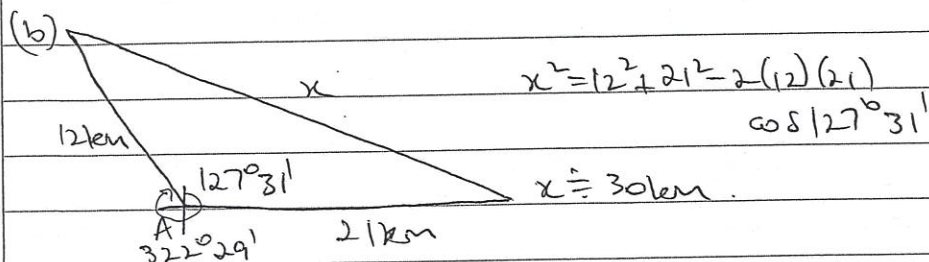
f) $\frac{\sin^3 \theta}{\cos \theta} + \sin \theta \cos \theta = \frac{\sin^3 \theta + \sin \theta \cos^2 \theta}{\cos \theta}$
 $= \frac{\sin \theta}{\cos \theta} (\sin^2 \theta + \cos^2 \theta) \frac{\sin \theta}{\sin^2 \theta + \cos^2 \theta}$
 $= \frac{\sin \theta}{\cos \theta} = \tan \theta$

11std

Start here for
Question Number:

12

$$\begin{aligned}
 (a) \quad h=9, A &= \frac{h}{3} [(y_1+y_2)+4(\text{odds})+2(\text{evens})] \\
 &= \frac{9}{3} [(12+18)+4(16+14)+2(14)] \\
 &= 534 \text{ m}^2
 \end{aligned}$$



$$(c) \quad y = x^3 - 6x^2 - 9x + 4$$

$$y' = 3x^2 - 12x - 9$$

$$= 3(x^2 - 4x - 3)$$

$$y' = 0, \text{ stat. pt.}$$

$$x^2 - 4x - 3 = 0$$

$$x = \frac{4 \pm \sqrt{(-4)^2 - 4(1)(-3)}}{2}$$

$$= \frac{4 \pm \sqrt{16+12}}{2}$$

$$= \frac{4 \pm \sqrt{28}}{2}$$

$$= 2 \pm \sqrt{7}$$

$$x = 4.65, y = -67.04 \text{ (min)}$$

$$x = -0.65, y = 7.04 \text{ (max)}$$

$$y' = 6x - 12$$

$$\text{when } x = 4.65, y'' = 15.87 > 0, \text{ min.}$$

$$\text{when } x = -0.65, y'' = -15.87 < 0, \text{ max}$$

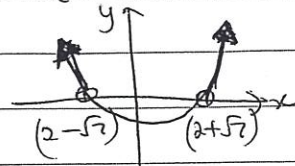
(iii) Increasing $y' > 0$, concave up $y'' > 0$.

$$x^2 - 4x - 3 > 0$$

$$6x - 12 > 0$$

$$6x > 12$$

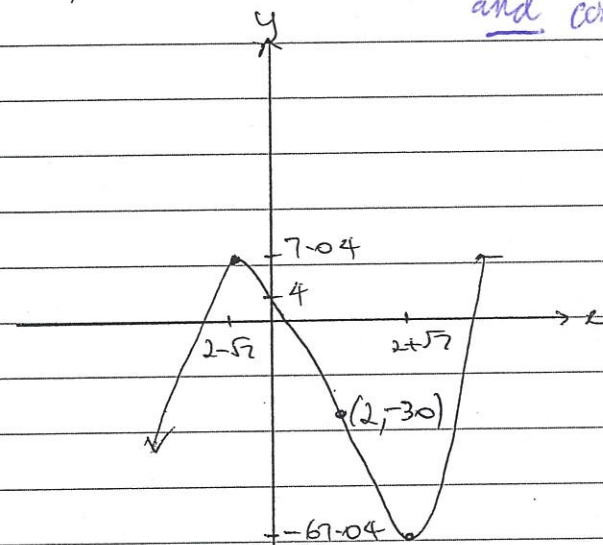
$$x > 2$$



$$\therefore x < 2 - \sqrt{7}, x > 2 + \sqrt{7}$$

$\therefore x > 2 + \sqrt{7}$ for increasing
and concave up.

(iv)



$$(d) \quad F(2,1) \quad V(3,1)$$

$$(i) \quad a=1$$

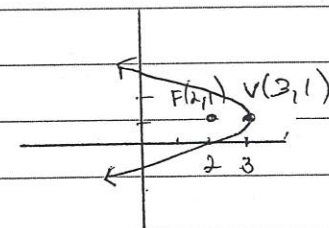
$$(ii) \quad x=4$$

$$(iii) \quad (y-k)^2 = -4(x-h)$$

$$(y-1)^2 = -4(x-3)$$

$$y^2 - 2y + 1 = -4x + 12$$

$$4x + y^2 - 2y - 11 = 0$$



Additional writing space on back page.



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Q13 a) $V = \pi \int_0^3 (10-3x)^2 dx$ or $\pi \int_0^3 (100-60x+9x^2) dx$

$$= \pi \left[100x - 30x^2 + 3x^3 \right]_0^3$$

$$= \pi \times 300 - 270 + 81$$

$$= 111\pi u^3$$

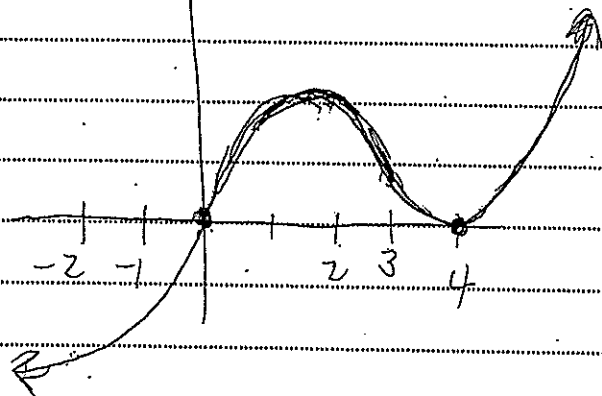
or

$$= \pi \left[\frac{-1}{9} - \left(\frac{-1000}{9} \right) \right]$$

$$= \frac{999\pi}{9} u^3$$

$$= 111\pi u^3$$

b)



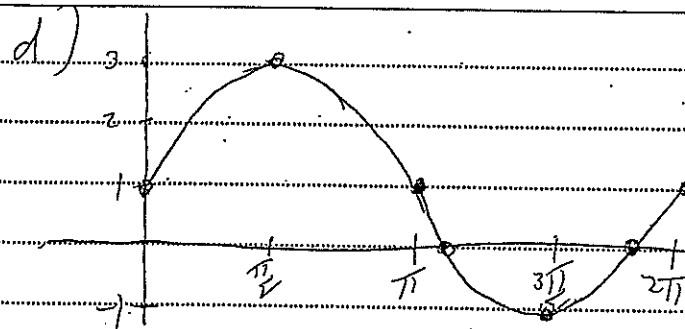
c) i) $\int (x+6)^{-\frac{1}{2}} dx = 2(x+6)^{\frac{1}{2}} + C$

ii) $\int_0^2 \frac{2x}{x^2+4} dx = \frac{1}{2} \ln(x^2+4) \Big|_0^2 = \frac{1}{2} (\ln 8 - \ln 4)$

$\frac{1}{2} \ln 2$



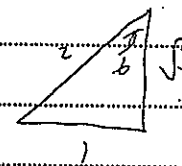
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ii) $1 + 2\sin \alpha = 0$

$$\sin \alpha = -\frac{1}{2}$$

$$\alpha = \frac{7\pi}{6}, \frac{11\pi}{6}$$



e) $16x^2 - (16a^2 - 24a + 9)$

$$16x^2 - (4a-3)^2$$

$$(4x-4a+3)(4x+4a-3)$$

Q1.4 $\sqrt{3}x - y + \sqrt{3} = 0$

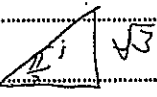
i) $A: y=0$ $\sqrt{3}x + \sqrt{3} = 0$

$x = -1$ $(-1, 0)$

$B: x=0$ $-y + \sqrt{3} = 0$

$y = \sqrt{3}$ $(0, \sqrt{3})$

ii) $m_{AB} = \frac{-\sqrt{3}}{-1} = \sqrt{3}$

iii) $\angle BAO$ $\tan \theta = \sqrt{3}$ 
 $\theta = \frac{\pi}{3}$ or 60°

IV) $AC: \frac{1}{2} AC \times AO \times \sin \frac{\pi}{3} = 3\sqrt{3}$

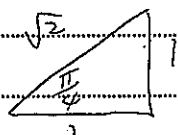
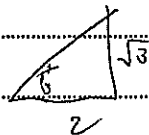
$\frac{1}{2} \times AC \times 1 \times \frac{\sqrt{3}}{2} = 3\sqrt{3}$

$\frac{\sqrt{3}}{4} AC = 3\sqrt{3}$

$AC = 12$

b) $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\pi}{3} + \operatorname{cosec} \frac{\pi}{4}$

$\sqrt{3} + \sqrt{2}$



$=$

c) $\log_e x = 2 \log_e 2x$

$x = 4x^2$

$4x^2 - x = 0$

$x(4x-1) = 0$

$x = 0, \frac{1}{4}$

but $x > 0$ for $\log_e x$

so $x = \frac{1}{4}$ is

the only solution

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Caringbah High School

Examination:

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Student Number

d) ~~$kx - kx$~~ $x^2 - (k+2)x + 4 = 0$

$\Delta = (k+2)^2 - 16 \geq 0$

$k^2 + 4k - 12 \geq 0$

$(k+6)(k-2) \geq 0$

$k \leq -6, k \geq 2$

e) $\frac{12+x}{24} = \frac{40}{12}$

$144 + 12x = 960$

$12x = 816$

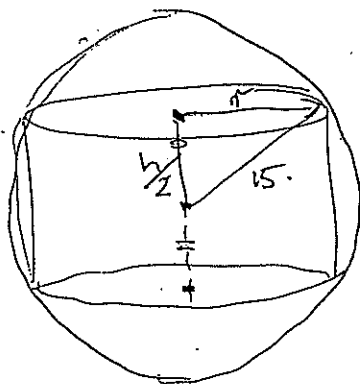
$x = 68$

f) $\lim_{x \rightarrow 3} \frac{(x-3)(x+2)}{x-3} = \lim_{x \rightarrow 3} x+2$
 $= 5$

11std



(a)



$$15^2 = r^2 + \left(\frac{h}{2}\right)^2$$

$$V = \pi r^2 h$$

$$225 = r^2 + \frac{h^2}{4}$$

$$V = \pi \left(225 - \frac{h^2}{4}\right) h$$

$$225 - \frac{h^2}{4} = r^2$$

$$V = 225\pi h - \frac{\pi h^3}{4}$$

$$(ii) V = 225\pi h - \frac{\pi h^3}{4}$$

$$\frac{dV}{dh} = 225\pi - \frac{3\pi h^2}{4}$$

$$\text{when } \frac{dV}{dh} = 0$$

$$\frac{3\pi h^2}{4} = 225\pi$$

$$3h^2 = 900$$

$$h^2 = 300$$

$$h = 10\sqrt{3}$$

$$\frac{d^2V}{dh^2} = -\frac{6\pi h}{4}$$

$\therefore$ Max value
when $h = 10\sqrt{3}$

$$b) 2x^2 + 3x - 4 = 0$$

$$i) x + \alpha = \frac{-3}{2}$$

$$ii) x^3 + \beta^3 = (x + \beta)(x^2 - x\beta + \beta^2)$$

$$= (x + \beta)(x + \beta)^2 - 3x\beta$$

$$= \frac{-3}{2} \left(\frac{9}{4} - 3 \times (-2) \right)$$

$$= \frac{-3}{2} \left(\frac{9}{4} + 6 \right)$$

$$= \frac{-3}{2} \times \frac{33}{4}$$

$$= \frac{-99}{8}$$

$$c) A_0 = 120000 \quad n = 120 \quad r = 0.005$$

$$A_0 = 120000 \times 1.005$$

$$= 120,600$$

$$ii) A_1 = 120600 - M$$

$$A_2 = 120600 \times 1.005 - M \times 1.005 - M$$

$$= 121203 - 2.005M$$

$$iii) A_3 = 120000 \times 1.005 - M(1 + 1.005 \times 1.005 - 1.005^2)$$

$$A_n = 120000 \times 1.005^n - M \left(\frac{1(1.005^n - 1)}{1.005 - 1} \right)$$

$$= 120000 \times 1.005^2 - 200M(1.005^2 - 1)$$

$$iv) M = \frac{120000 \times 1.005^{20}}{200(1.005^{20} - 1)}$$

$$= \$1332.25$$

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Student Number

Examination:

$$Q16) i) V = V_0 e^{-kt}$$

$$\frac{dV}{dt} = -k(V_0 e^{-kt})$$

$$= -kV$$

$$ii) \text{ when } t = 0 \quad V = 1000$$

$$\therefore V = 1000 e^{-kt}$$

$$\text{when } t = 40 \quad V = 800$$

$$800 = 1000 e^{-40k}$$

$$e^{-40k} = \frac{4}{5}$$

$$-40k = \ln \frac{4}{5}$$

$$k = \frac{-\ln \frac{4}{5}}{40}$$

$$V = 1000 e^{-60 \left(\frac{-\ln \frac{4}{5}}{40} \right)}$$

$$= 716L$$

$$iii) 10 = 1000 e^{-\left(\frac{\ln \frac{4}{5}}{40} \right) t}$$

$$0.01 = e^{-\frac{\ln \frac{4}{5}}{40} t}$$

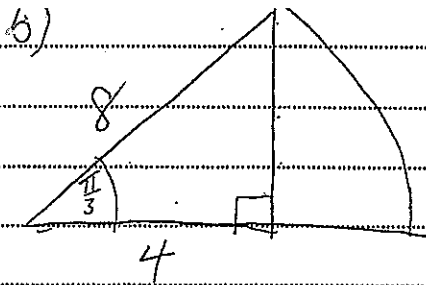
$$\frac{\ln \frac{4}{5}}{40} t = \ln 0.01$$

$$t = \frac{40 \ln 0.01}{\ln \frac{4}{5}}$$

$$= 13 \text{ hrs } 46 \text{ min.}$$

$$= 826 \text{ min}$$

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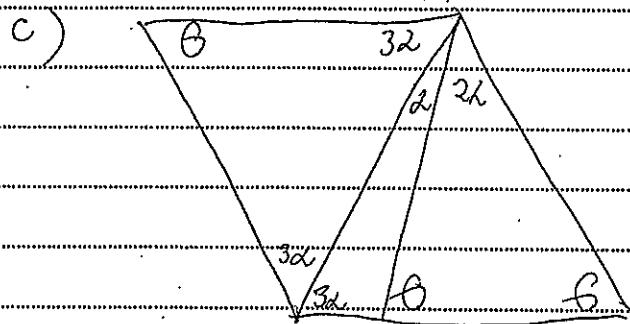
$$\frac{1}{2} \times 4 \times 8 \times \sin \frac{\pi}{3}$$

$$= \frac{16\sqrt{3}}{2} = 8\sqrt{3}$$

$$\frac{1}{2} \times 8 \times 8 \times \sin \frac{\pi}{3}$$

$$= \frac{32\sqrt{3}}{2} = 16\sqrt{3}$$

$$\text{Area} = \frac{32\sqrt{3}}{3} - 8\sqrt{3}$$



$$12\alpha + 2\beta = 360$$

$$2\alpha + 2\beta = 180$$

$$10\alpha = 180$$

$$\alpha = 18^\circ$$

$$36 + 2\beta = 180$$

$$\beta = 72^\circ$$

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Examination:.....

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d) $\int \frac{x \log x - x}{dx}$

$u = x$ $v = \log x$
 $du = 1$ $dv = \frac{1}{x}$

$$= x \times \frac{1}{x} + \log x - 1$$

$$= 1 + \log x - 1$$

$$= \log x$$

$$\int \log(x^2) dx = 2 \int \log x dx$$

$$= 2(x \log x - x) + C$$

11) Area = $(2 \log 5) \times 5 - 2 \int \log(x^2) dx$

$$= 10 \log 5 - 2 \left[x \log x - x \right]_1^5$$

$$= 10 \log 5 - 2 \left[5 \log 5 - 5 - (1 \log 1 - 1) \right]$$

$$10 \log 5 - (10 \log 5 - 4)$$

$$= 4$$

11std

Q. 16 d ii) Alternative method:

$$\int_0^{\ln 25} \log_e(x^2) dy$$

$$y = \log_e x^2$$

$$y = 2 \log_e x$$

$$\log_e x = \frac{y}{2}$$

$$x = e^{\frac{y}{2}}$$

$$\stackrel{\ln 25}{=} \int_0^{\ln 25} e^{\frac{y}{2}} dy$$

$$= 2e^{\frac{y}{2}} \Big|_0^{\ln 25}$$

$$= 2 \left[e^{\frac{1}{2} \ln 25} - e^0 \right]$$

$$= 2 \left[e^{\ln 5} - 1 \right]$$

$$= 2 \left[4 \right]$$

$$= 8$$